

for refusal prefixes in these token positions. This is plausible, as we note that a full fine-tuning on SQL Create Context and Samsum with more than 600 gradient steps results in an increase of ASR from 1.5% to 14.9% and 25.5% respectively, but the ASR is already at 13.6% and 22.1% after the initial 10 gradient steps. This suggests that the most significant safety regression exactly occurs during these early steps when the gradient norm for the initial tokens is excessively large.

C Ablation Studies on Fine-tuning Attack Experiments

This section presents more in-depth ablation studies to supplement our studies in Section 4 and Table 3 there. We also supplement Section 3.2 by presenting results (Table 6) of fine-tuning attacks on the augmented model that we build in Section 3.

Biased Constrains on The Early Tokens Are Important. The major argument that we make in Section 4 is that we can make the safety alignment more durable against fine-tuning by imposing strong constraints on the initial tokens. Therefore, we set a larger β_t to impose stronger constraints in early tokens while only setting a very weak β_t for later tokens. Our results in table 3 indeed verify the improved safety. To further support that this improvement is indeed due to the biased constraints on the early tokens, we perform an ablation where all β_t are set to a uniform value. Results are presented in Table 4. As shown, if we set the same small $\beta = 0.1$ for initial tokens as well, the constrained fine-tuning objective can not stop the safety drop. While if we set the large $\beta = 2.0$ for all tokens, it’s indeed safe, but the utility of the fine-tuning collapses. Similarly, $\beta = 0.5$ for all tokens neither achieve optimal safety, and the utility is worse than the biased configurations we use in Table 3.

Table 4: Ablation on β_t in Eqn 3. (Fine-tuning Llama-2-7B-Chat)

Datasets		Initial	Standard SFT	Constrained SFT (biased β_t)	Constrained SFT (uniform $\beta = 0.1$)	Constrained SFT (uniform $\beta = 0.5$)	Constrained SFT (uniform $\beta = 2.0$)
<i>Against Fine-tuning Attacks</i>							
Harmful Examples	ASR	1.5%	88.9%	4.6%	86.2%	7.2%	0.5%
Identity Shifting	ASR	0%	79.5%	8.1%	41.6%	17.1%	3.4%
Backdoor	ASR (w/o trigger)	1.5%	7.6%	1.9%	3.5%	1.8%	1.2%
Poisoning	ASR (w/ trigger)	1.7%	90.9%	10.9%	74.4%	24.3%	1.4%
<i>Fine-tuning with Normal Downstream Datasets</i>							
Samsum	ASR	1.5%	23.4%	3.2%	3.9%	3.5%	2.4%
	Utility	25.5%	51.7%	50.1%	51.7%	49.8%	42.5%
SQL Create Context	ASR	1.5%	15.4%	3.2%	3.3%	2.2%	2.6%
	Utility	14.9%	99.1%	98.5%	99.1%	98.6%	92.6%
GSM8k	ASR	1.5%	3.3%	2.0%	4.0%	1.5%	2.0%
	Utility	25.5%	41.7%	37.4%	39.4%	34.8%	2.1%

Ablation on The Effects of Warmup Steps. As we have also noted in Appendix A.5.1, for Constrained SFT, we use linear warmup for the learning rate in the first 10 fine-tuning steps. This warmup makes sure the constraints imposed by the sigmoid function are gently initialized — note that, at the start of the fine-tuning, the log ratio $\log \pi_\theta / \pi_{\text{aligned}}$ in Eqn 3 is equal to 0. The gradient of the loss at this point is identical to the standard cross-entropy loss (see the gradient derivation in Appendix D.2). Therefore, in stochastic gradient descent, there is a risk that early gradients will already break the alignment without respecting the constraints. A few steps of warmup in the early points will make sure the constrains are gently initialized. We present an ablation study on the effect of these 10 steps of warmup in Table 5. We can summarize two key takeaways: 1) the warmup steps are indeed useful to make the constrained SFT to be perform consistently safer; 2) the safety improvement is not solely coming from the warmup steps but mostly from the constrained SFT optimization objective we design.

Fine-tuning Attacks on The Augmented Model We Build in Section 3. Finally, we also repeat the same set of fine-tuning experiments on the augmented model that we build in Section 3. Results are presented in Table 6. By comparing the results of SFT in Table 6 and Table 5, we can see the augmented model is generally more robust in multiple fine-tuning cases compared with the non-augmented model. And the constrained fine-tuning objective can also be applied to it, though we didn’t observe consistently better results when the two techniques are combined.

Table 5: Ablation on The Effects of The 10 Warmup Steps. (Fine-tuning Llama-2-7B-Chat)

Datasets		Initial	Standard SFT	Standard SFT (with warmup)	Constrained SFT	Constrained SFT (with warmup)
<i>Against Fine-tuning Attacks</i>						
Harmful Examples	ASR	1.5%	88.9%	89.4%	29.1%	4.6%
Identity Shifting	ASR	0%	79.5%	44.8%	69.6%	8.1%
Backdoor Poisoning	ASR (w/o trigger)	1.5%	7.6%	2.7%	2.7%	1.9%
	ASR (w/ trigger)	1.7%	90.9%	80.5%	9.7%	10.9%
<i>Fine-tuning with Normal Downstream Datasets</i>						
Samsun	ASR	1.5%	23.4%	3.8%	23.1%	3.2%
	Utility	25.5%	51.7%	51.9%	50.2%	50.1%
SQL Create Context	ASR	1.5%	15.4%	3.3%	2.0%	3.2%
	Utility	14.9%	99.1%	99.1%	98.6%	98.5%
GSM8k	ASR	1.5%	3.3%	2.9%	3.1%	2.0%
	Utility	25.5%	41.7%	41.6%	37.2%	37.4%

Table 6: Fine-tuning Llama-2-7B-Chat-Augmented That We Built in Section 3. Refer to Table 5 for the results on the non-augmented counterpart, i.e., Llama-2-7B-Chat.

Datasets		Standard SFT	Standard SFT (with warmup)	Constrained SFT	Constrained SFT (with warmup)
<i>Against Fine-tuning Attacks</i>					
Harmful Examples	ASR	55.2%	51.5%	9.4%	5.2%
Identity Shifting	ASR	53.9%	37.0%	28.8%	3.0%
Backdoor Poisoning	ASR (w/o trigger)	3.9%	2.7%	1.8%	0.9%
	ASR (w/ trigger)	80.0%	83.6%	16.4%	12.7%
<i>Fine-tuning with Normal Downstream Datasets</i>					
Samsun	ASR	2.1%	0.6%	2.1%	1.2%
	Utility	52.4%	51.9%	50.4%	50.1%
SQL Create Context	ASR	3.8%	1.5%	2.0%	0.9%
	Utility	99.0%	99.1%	98.4%	98.5%
GSM8k	ASR	0.9%	0.9%	0.3%	0.3%
	Utility	42.0%	41.2%	36.5%	36.9%

D Interpretation of Our Constrained Fine-tuning Objective

In this section, we provide detailed interpretation for our constrained fine-tuning objective in Section 4. Recall that our fine-tuning objective is defined as:

$$\mathcal{L}(\theta) := \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} - \sum_{t=1}^{|\mathbf{y}|} \frac{2}{\beta_t} \log \left[\sigma \left(\beta_t \log \frac{\pi_{\theta}(y_t | \mathbf{x}, \mathbf{y}_{<t})}{\pi_{\text{aligned}}(y_t | \mathbf{x}, \mathbf{y}_{<t})} \right) \right]. \quad (6)$$

Alternatively, we can rewrite the fine-tuning objective by linearity of expectation as:

$$\mathcal{L}(\theta) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \sum_{t \geq 1} -\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \frac{2}{\beta_t} \log \left[\sigma \left(\beta_t \log \frac{\pi_{\theta}(y_t | \mathbf{x}, \mathbf{y}_{<t})}{\pi_{\text{aligned}}(y_t | \mathbf{x}, \mathbf{y}_{<t})} \right) \right] \quad (7)$$

$$= \sum_{t \geq 1} \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} -\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \frac{2}{\beta_t} \log \left[\sigma \left(\beta_t \log \frac{\pi_{\theta}(y_t | \mathbf{x}, \mathbf{y}_{<t})}{\pi_{\text{aligned}}(y_t | \mathbf{x}, \mathbf{y}_{<t})} \right) \right] \quad (8)$$

$$= \sum_{t \geq 1} \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \mathbb{1}_{\{t \leq |\mathbf{y}|\}} \frac{2}{\beta_t} S \left(-\beta_t \log \frac{\pi_{\theta}(y_t | \mathbf{x}, \mathbf{y}_{<t})}{\pi_{\text{aligned}}(y_t | \mathbf{x}, \mathbf{y}_{<t})} \right), \quad (9)$$

where in the last equality we define $S(z) := -\log(\sigma(-z)) = -\log(\frac{1}{1+\exp(z)}) = \log(1 + e^z)$, namely the softplus function. Therefore, we can split $\mathcal{L}(\theta)$ into a sum of token-wise losses:

$$\mathcal{L}(\theta) = \sum_{t \geq 1} \ell_t(\theta), \text{ where } \ell_t(\theta) := \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \mathbb{1}_{\{t \leq |\mathbf{y}|\}} \frac{2}{\beta_t} S \left(\beta_t \Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t) \right), \quad (10)$$

$$\Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t) := \log \pi_{\text{aligned}}(y_t | \mathbf{x}, \mathbf{y}_{<t}) - \log \pi_{\theta}(y_t | \mathbf{x}, \mathbf{y}_{<t}). \quad (11)$$

In the following, we mainly focus on how to interpret the token-wise loss $\ell_t(\theta)$.

D.1 Limiting Behaviors

D.1.1 Small β_t

When β_t is small, we have the following theorem showing that $\ell_t(\theta)$ in Eqn 10 is **approximately the cross-entropy loss** at position t , up to a constant.

Theorem 1. *For a given θ , as $\beta_t \rightarrow 0$, we have*

$$\ell_t(\theta) - C(\beta_t) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \left[-\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \cdot \log \pi_\theta(y_t \mid \mathbf{x}, \mathbf{y}_{<t}) \right] + O(\beta_t), \quad (12)$$

where

$$C(\beta_t) := \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \left[\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \left(\frac{2}{\beta_t} \log 2 + \log \pi_{\text{aligned}}(y_t \mid \mathbf{x}, \mathbf{y}_{<t}) \right) \right], \quad (13)$$

which is a bias term that is constant with respect to θ .

Proof. Recall that $\ell_t(\theta) := \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \mathbb{1}_{\{t \leq |\mathbf{y}|\}} \frac{2}{\beta_t} S(\beta_t \Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t))$ and $S(z) := \log(1 + e^z)$ is the softplus function [41]. By Taylor expansion, it holds for all z that $S(\beta_t z) = S(0) + S'(0)\beta_t z + S''(\xi\beta_t z)\beta_t^2 z^2$ for some $\xi \in (0, 1)$ depending on z . Since $S(0) = \log 2$, $S'(0) = \frac{1}{2}$ and $S''(x) \in [0, 1/4]$ for all x , we have $|S(\beta_t z) - (\log 2 + \frac{1}{2}\beta_t z)| \leq \frac{1}{4}\beta_t^2 z^2$. Dividing both sides by $\beta_t/2$, we get

$$\left| \frac{2}{\beta_t} S(\beta_t z) - \left(\frac{2}{\beta_t} \log 2 + z \right) \right| \leq \frac{1}{2} \beta_t z^2. \quad (14)$$

Then for our token-wise loss $\ell_t(\theta)$, we have

$$\left| \ell_t(\theta) - \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \left[\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \left(\frac{2}{\beta_t} \log 2 + \Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t) \right) \right] \right| \leq \frac{1}{2} \beta_t \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \mathbb{1}_{\{t \leq |\mathbf{y}|\}} \Delta_t^2(\mathbf{x}, \mathbf{y}_{<t}, y_t) = O(\beta_t).$$

Recall that $\Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t) := \log \pi_{\text{aligned}}(y_t \mid \mathbf{x}, \mathbf{y}_{<t}) - \log \pi_\theta(y_t \mid \mathbf{x}, \mathbf{y}_{<t})$ as per Eqn 11. We can thus have

$$\begin{aligned} \ell_t(\theta) - \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \left[\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \left(\frac{2}{\beta_t} \log 2 + \log \pi_{\text{aligned}}(y_t \mid \mathbf{x}, \mathbf{y}_{<t}) \right) \right] \\ = - \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \left[\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \log \pi_\theta(y_t \mid \mathbf{x}, \mathbf{y}_{<t}) \right] + O(\beta_t). \end{aligned} \quad (15)$$

Noting that $C(\beta_t) := \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} \left[\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \left(\frac{2}{\beta_t} \log 2 + \log \pi_{\text{aligned}}(y_t \mid \mathbf{x}, \mathbf{y}_{<t}) \right) \right]$, we can complete the proof. $\square$

D.1.2 Large β_t

When β_t is large, we have the following theorem showing that $\ell_t(\theta)$ can be approximated by $\tilde{\ell}_t(\theta) := \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} [\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \max\{\Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t), 0\}]$.

Theorem 2. *For a given θ , as $\beta_t \rightarrow +\infty$, we have*

$$\ell_t(\theta) = 2\tilde{\ell}_t(\theta) + O(\beta_t^{-1}),$$

where

$$\tilde{\ell}_t(\theta) := \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D} [\mathbb{1}_{\{t \leq |\mathbf{y}|\}} \max\{\Delta_t(\mathbf{x}, \mathbf{y}_{<t}, y_t), 0\}].$$

Proof. First, we note the following identity.

$$S(z) = \log(1 + e^z) = \max\{z, 0\} + \log((1 + e^z)e^{-\max\{z, 0\}}) = \max\{z, 0\} + \log(1 + e^{-|z|}). \quad (16)$$

This implies that

$$\left| \frac{2}{\beta_t} S(\beta_t z) - 2 \cdot \max\{z, 0\} \right| = \frac{2}{\beta_t} \log(1 + e^{-\beta_t |z|}) \leq \frac{2}{\beta_t} e^{-\beta_t |z|}. \quad (17)$$